EEE240 Fundamentals of Digital Logic Design

Chapter no. 4

Dr. Ghufran Shafiq

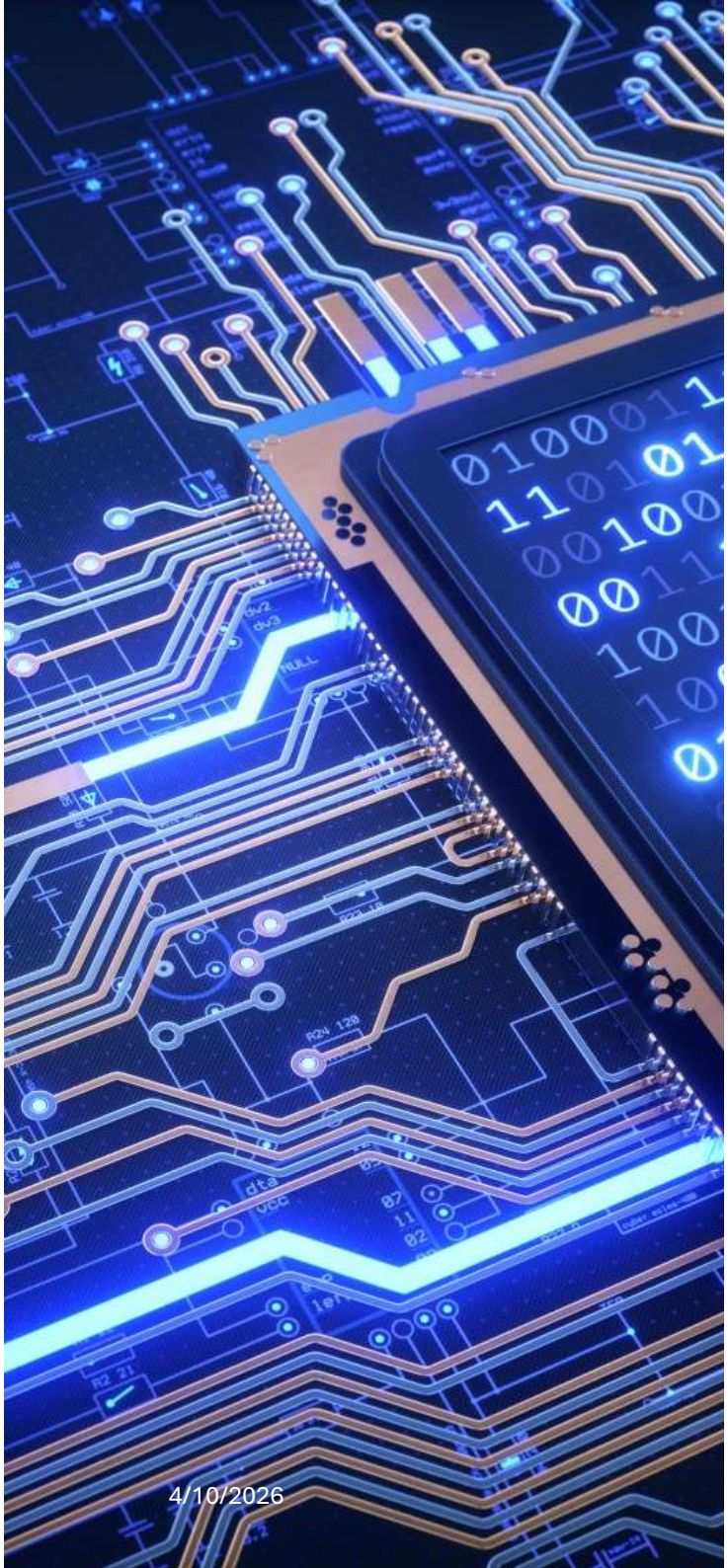
Assistant Professor

Department of Electrical Engineering

COMSATS University Islamabad

Review

- What is the utility of k Maps?
- For a 6 variable k Map, how many square boxes will be required?
- What is the rule for two adjacent squares in k-maps?
- What is the rule for combining the squares? i.e. rule for size of any set?
- What are implicants, prime implicants and essential prime implicants? What is the rule for getting the minimized expression?
- How do we get the expression for a complement of a function from the k-map?
- What is the advantage of considering don’t care condition?

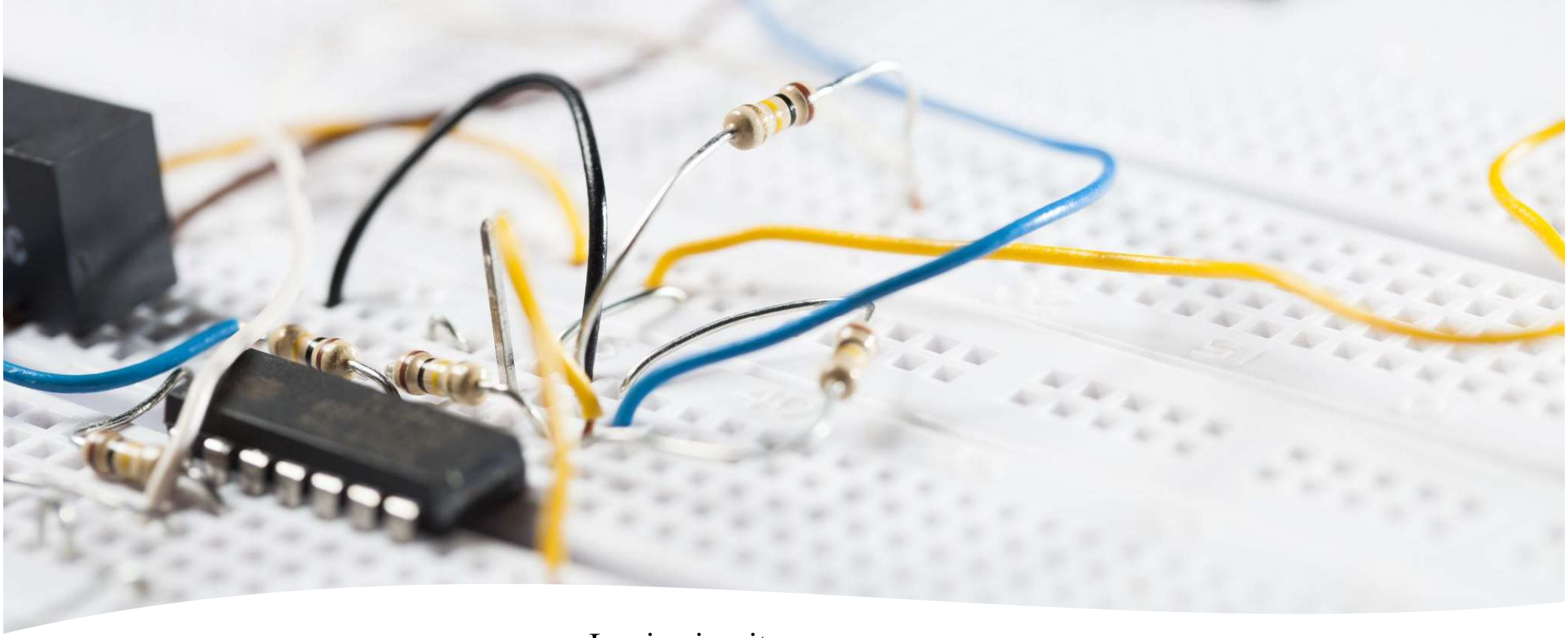


Outline

- Introduction to combinational logic
- Combinational circuits
- Analysis procedure
- Design procedure
- Binary adder-subtractor
- Decimal adder
- Binary multiplier
- Magnitude comparator
- Decoders
- Encoders
- Multiplexers

Must Read

Chapter No. 4: Combinational Logic



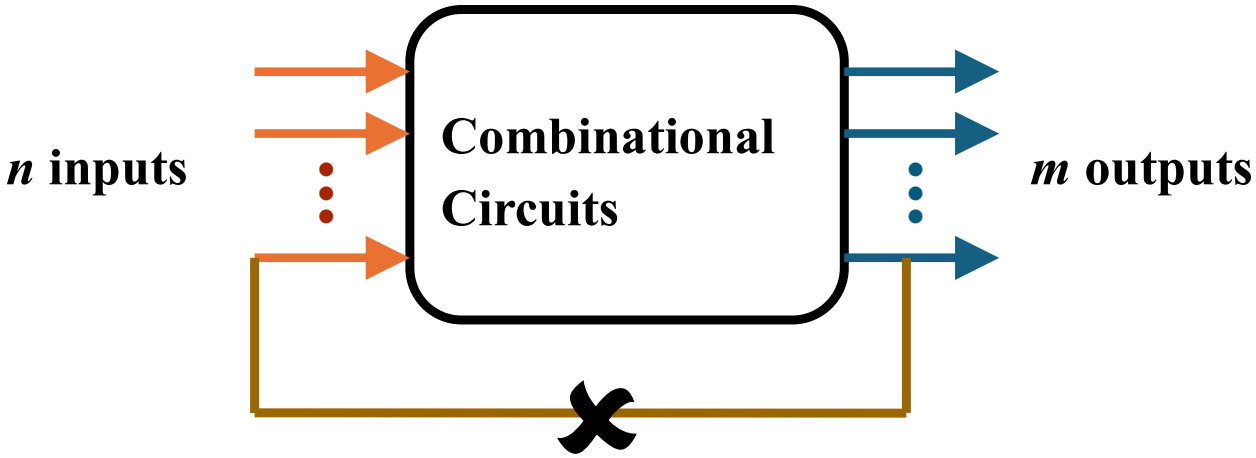
Introduction

- Logic circuits
  - Combinational
    - Only logic gates
    - Output depends only on present combination of inputs
    - Can be specified logically by a set of Boolean functions
  - Sequential
    - Logic gates with storage elements
    - Outputs are function of inputs and state of the storage elements

Combinational Circuits

- Output is function of input only

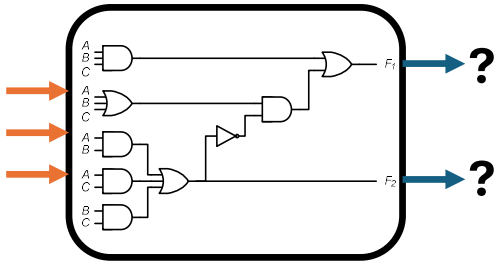
  i.e. no feedback



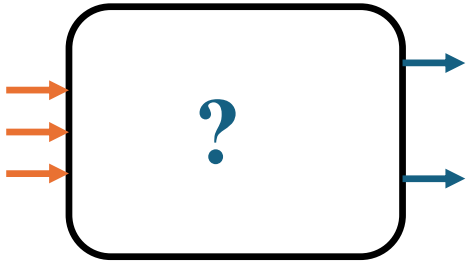
When input changes, output may change (after a delay)

Combinational Circuits

- Analysis
  - Given a circuit, find out its *function*
  - Function may be expressed as:
    - Boolean function
    - Truth table

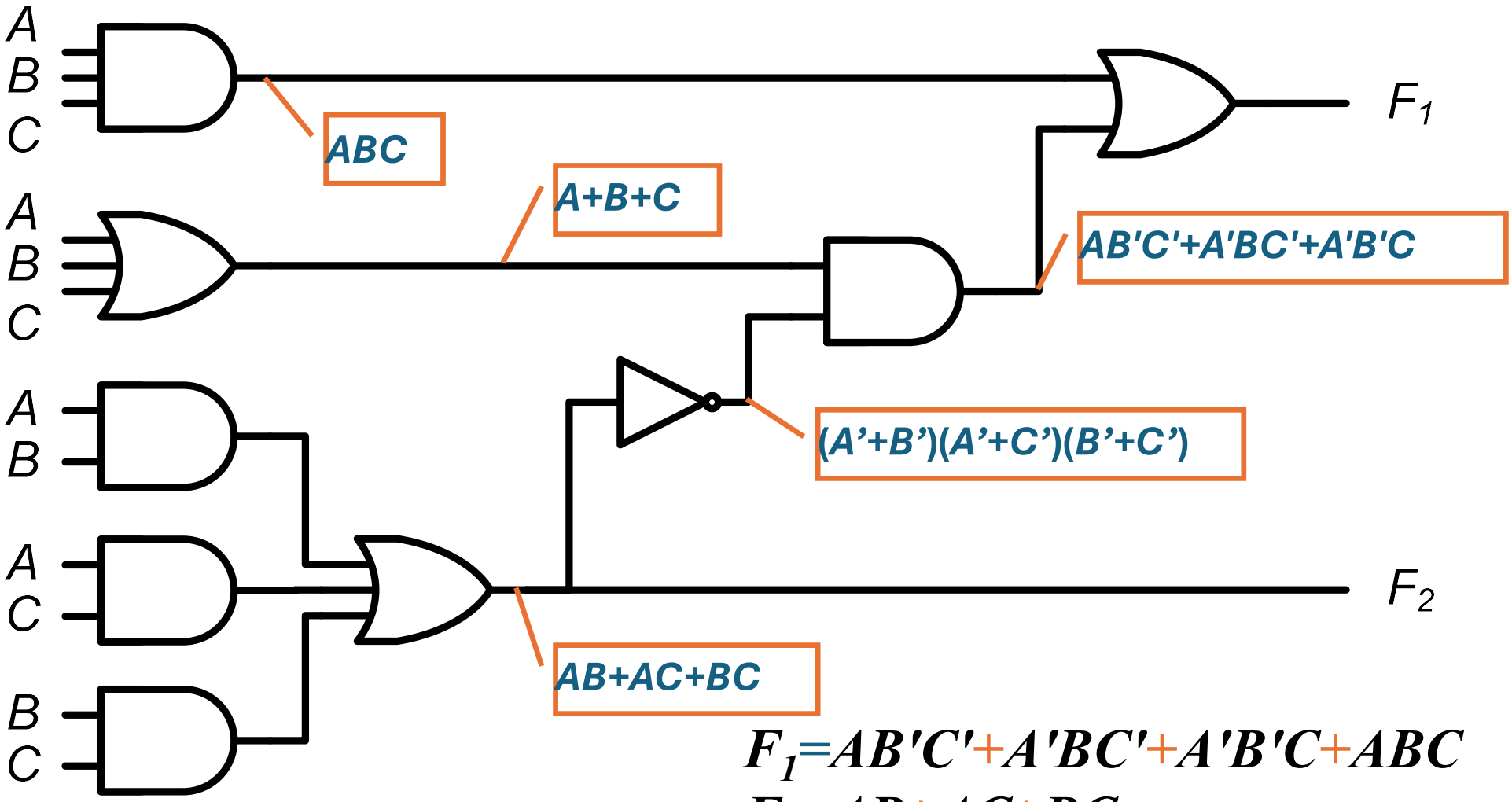


- Design
  - Given a desired function, determine its *circuit*
  - Function may be expressed as:
    - Boolean function
    - Truth table



Analysis Procedure

- Boolean Expression Approach

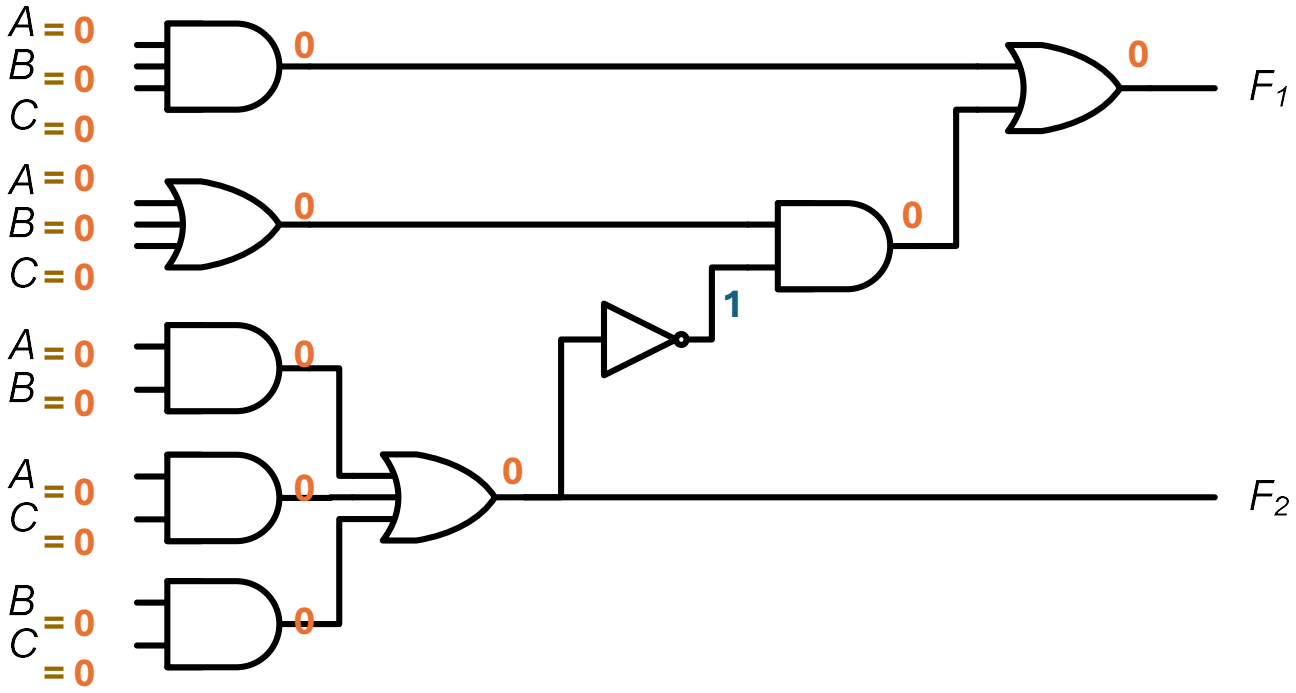


$F_1 = AB'C' + A'BC' + A'B'C + ABC$

$F_2 = AB + AC + BC$

Analysis Procedure

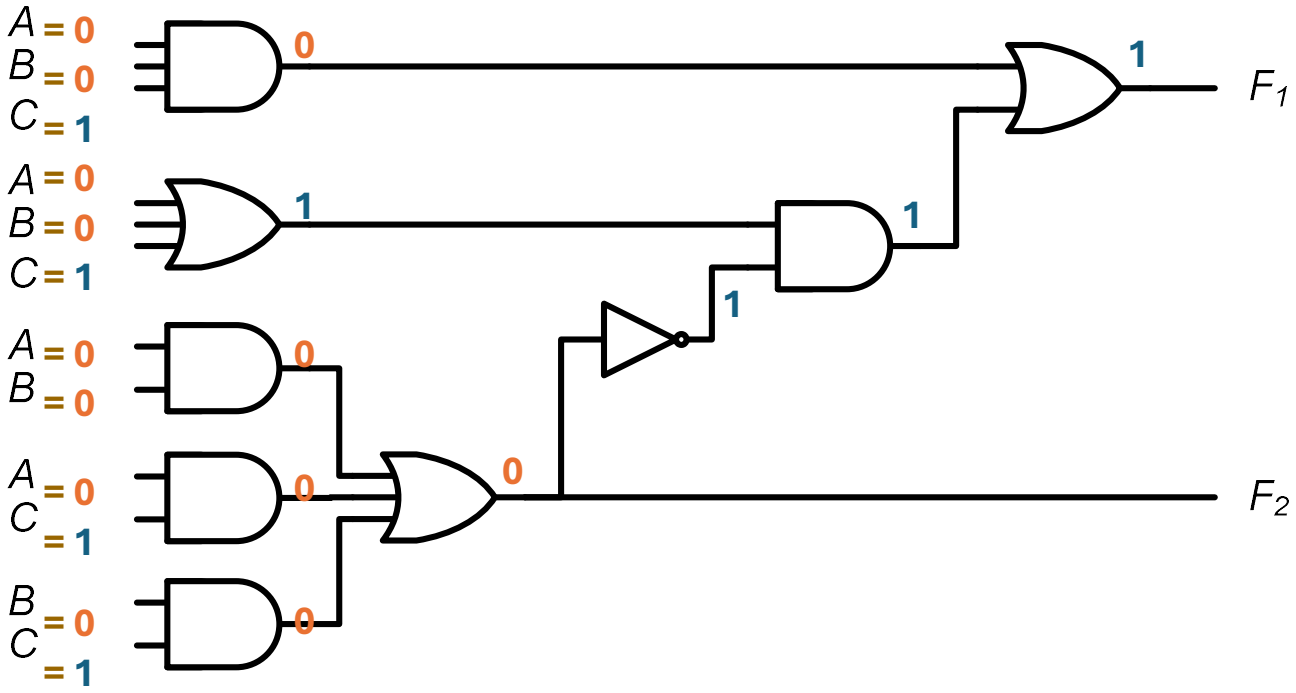
- Truth Table Approach



A	B	C	F_1	F_2
0	0	0	0	0

Analysis Procedure

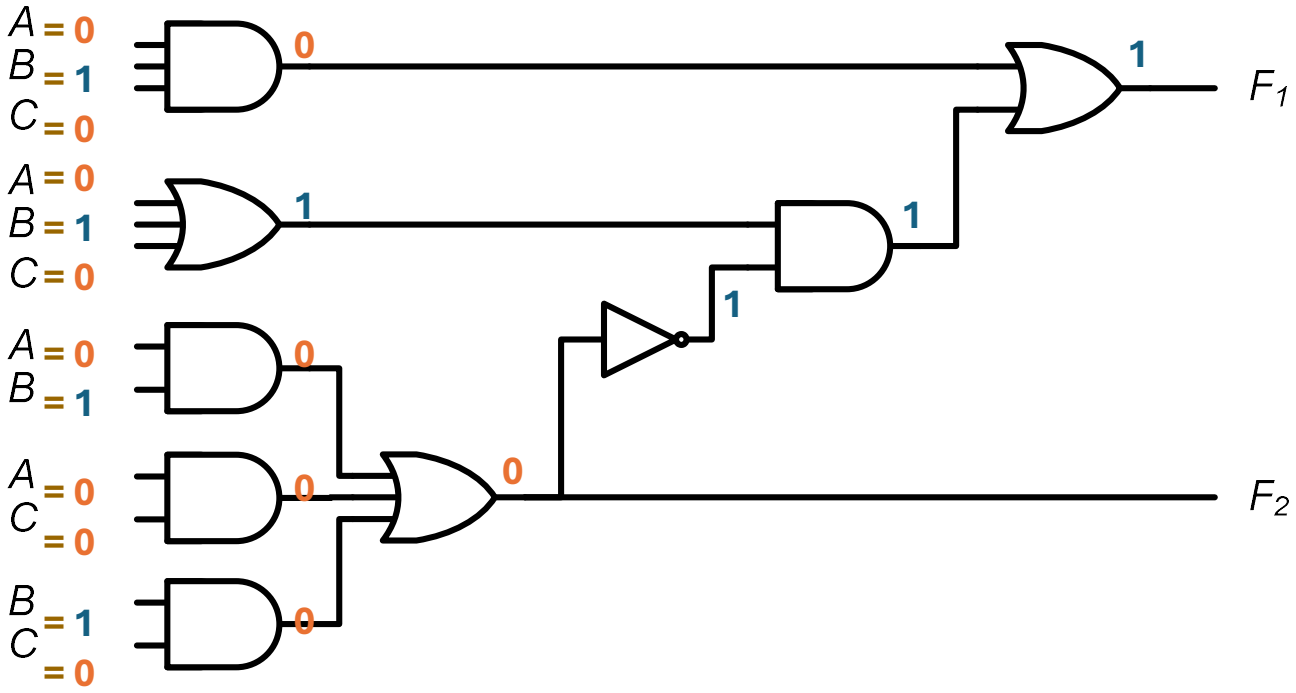
- Truth Table Approach



A	B	C	F_1	F_2
0	0	0	0	0
0	0	1	1	0

Analysis Procedure

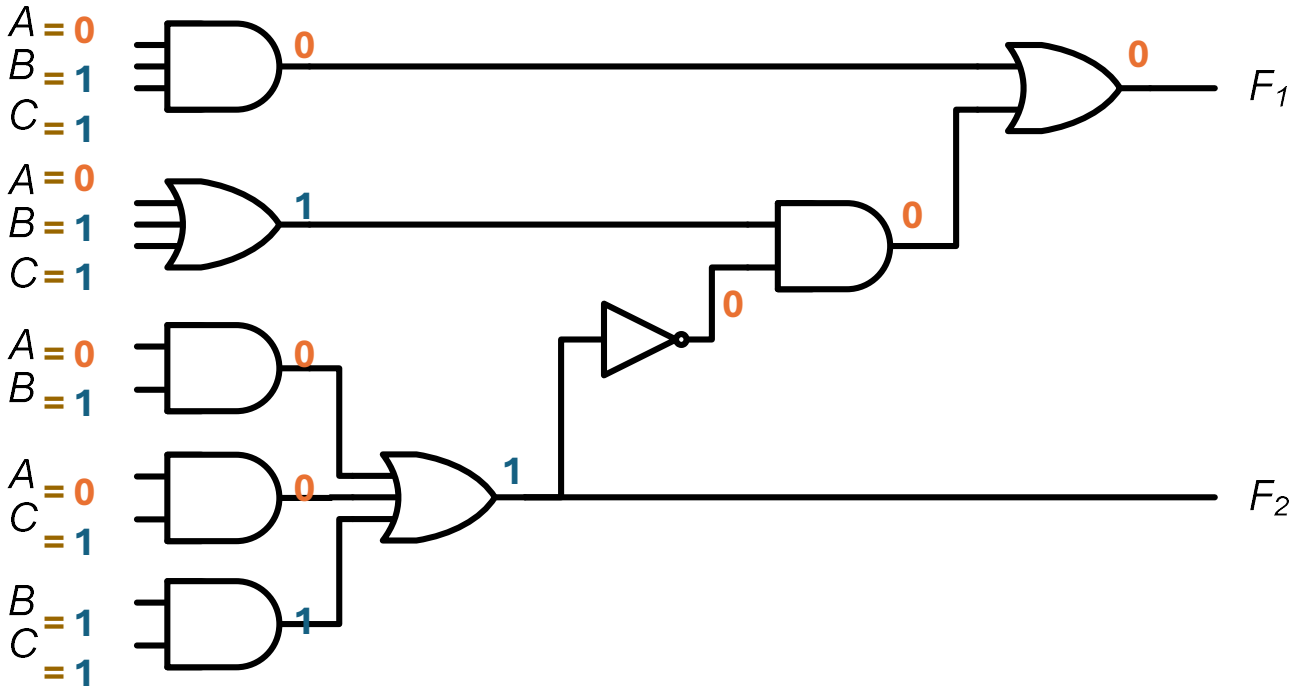
- Truth Table Approach



A	B	C	F_1	F_2
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0

Analysis Procedure

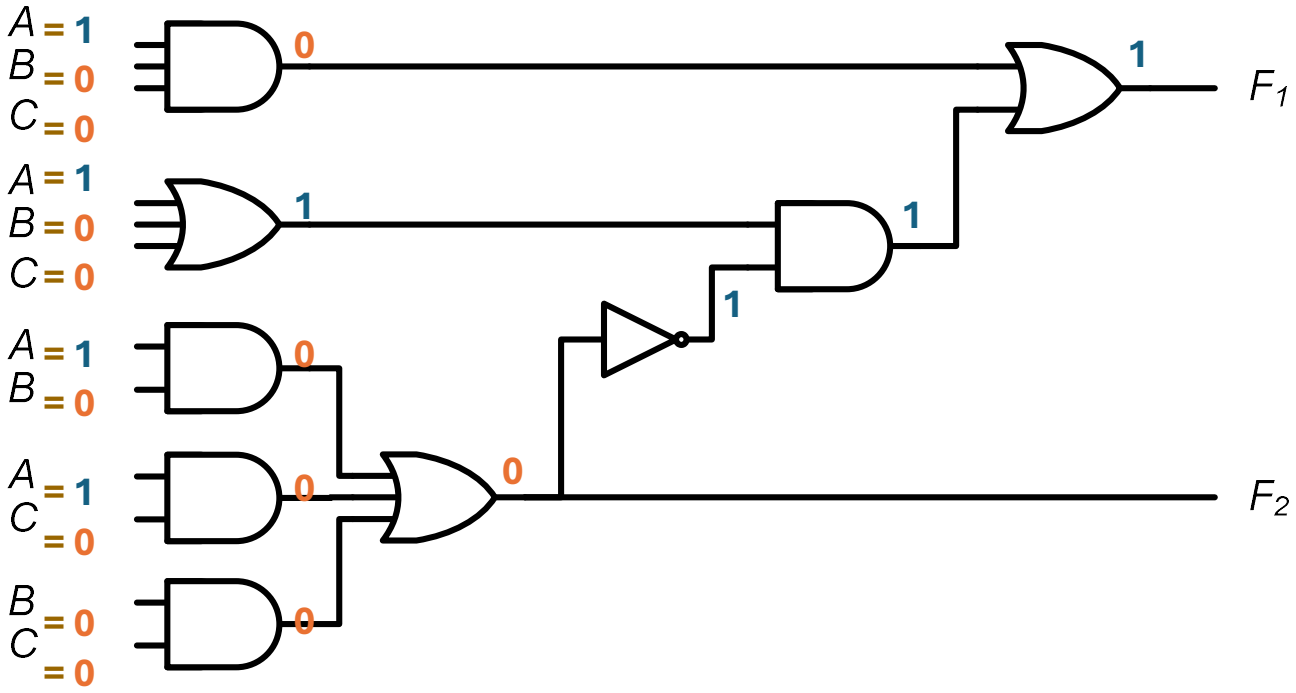
- Truth Table Approach



A	B	C	F_1	F_2
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1

Analysis Procedure

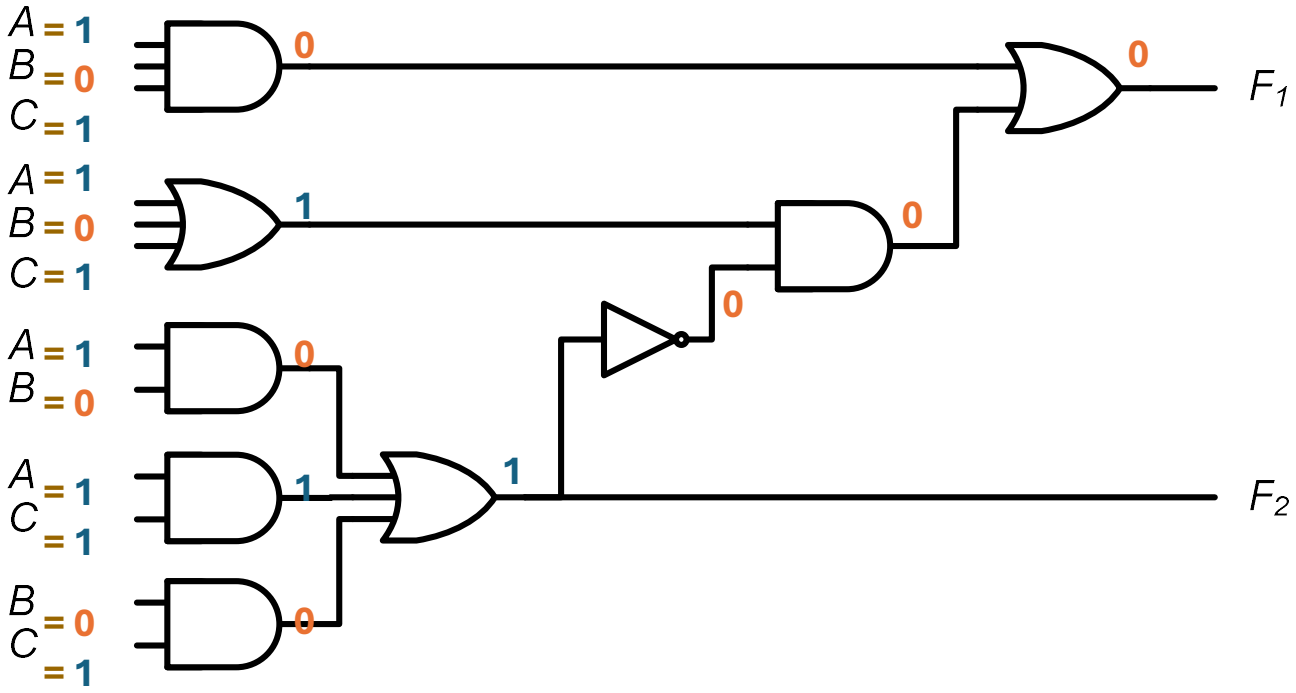
- Truth Table Approach



A	B	C	F_1	F_2
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0

Analysis Procedure

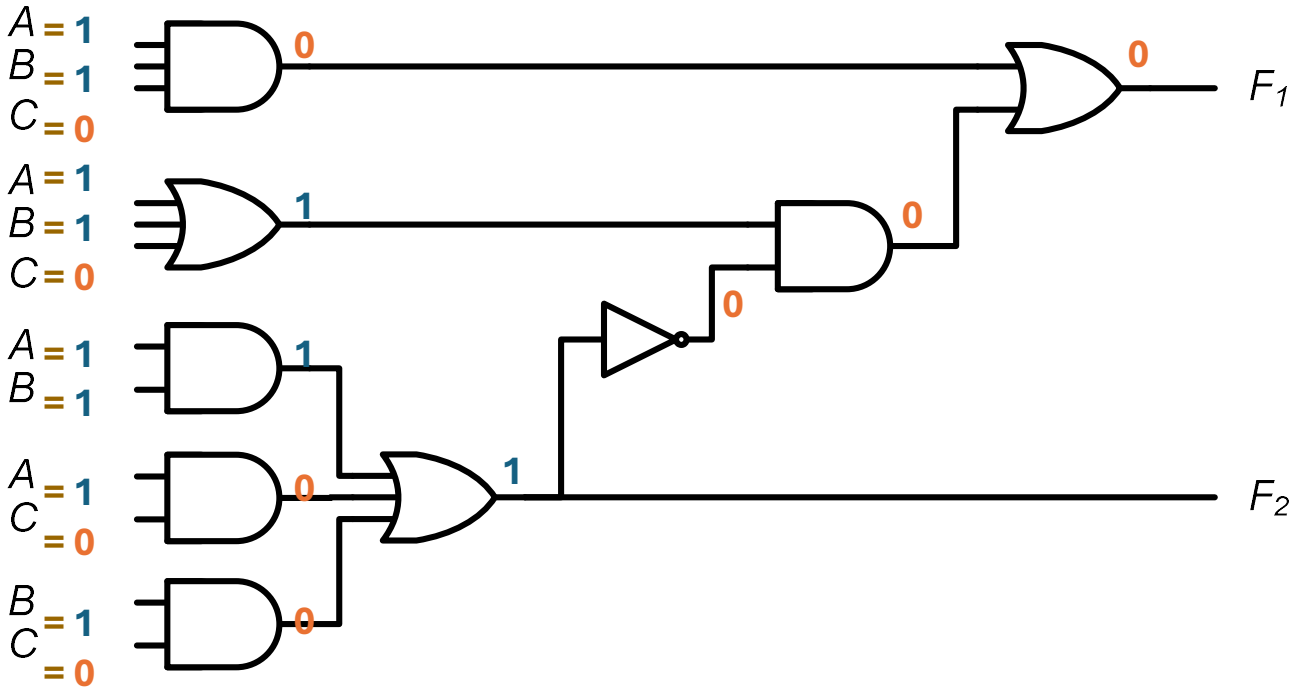
- Truth Table Approach



A	B	C	F_1	F_2
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1

Analysis Procedure

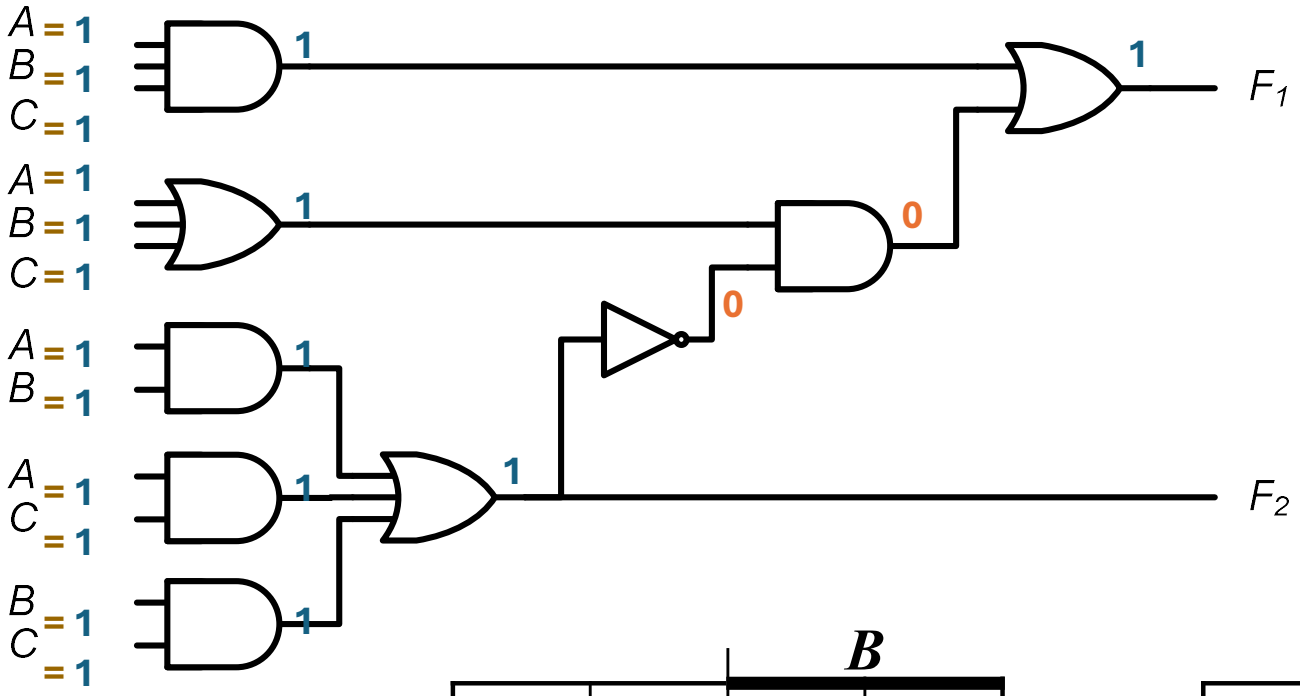
- Truth Table Approach



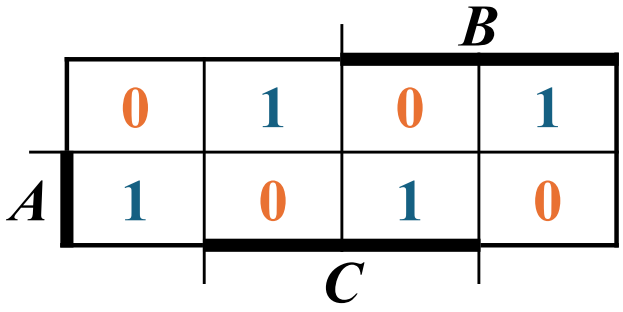
A	B	C	F_1	F_2
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1

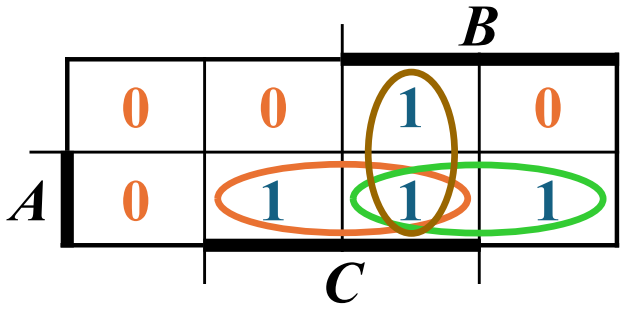
Analysis Procedure

- Truth Table Approach



A B C	F_1	F_2
0 0 0	0	0
0 0 1	1	0
0 1 0	1	0
0 1 1	0	1
1 0 0	1	0
1 0 1	0	1
1 1 0	0	1
1 1 1	1	1





$F_1 = AB'C' + A'BC' + A'B'C + ABC$

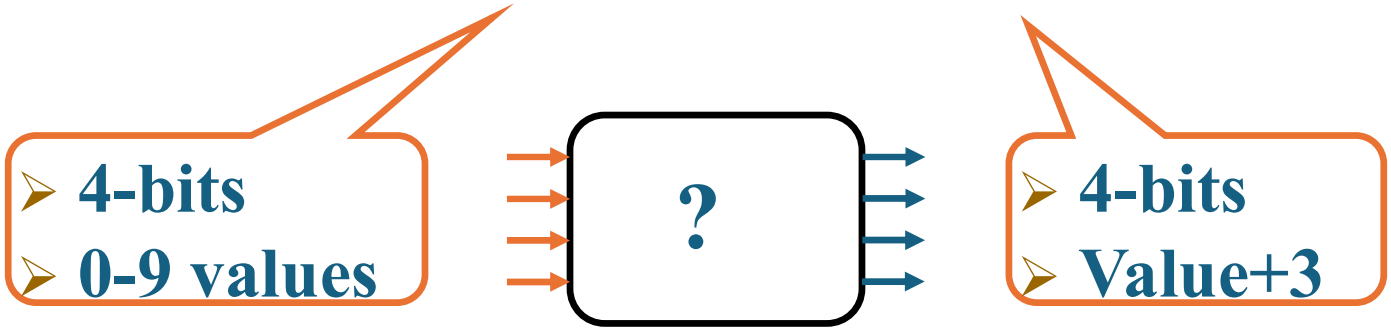
$F_2 = AB + AC + BC$

Design Procedure

- Given a problem statement:
  - Determine the number of *inputs* and *outputs*
  - Derive the truth table
  - Simplify the Boolean expression for each output
  - Produce the required circuit

Example:

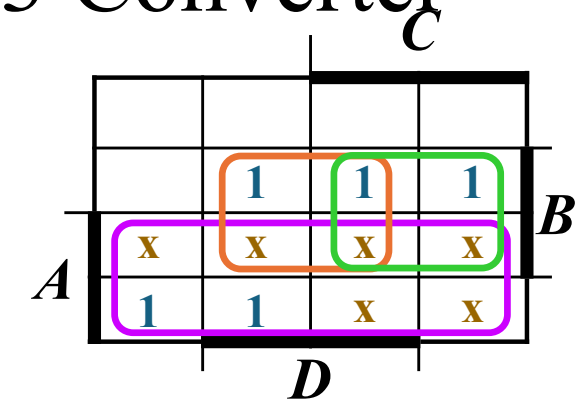
Design a circuit to convert a “BCD” code to “Excess 3” code



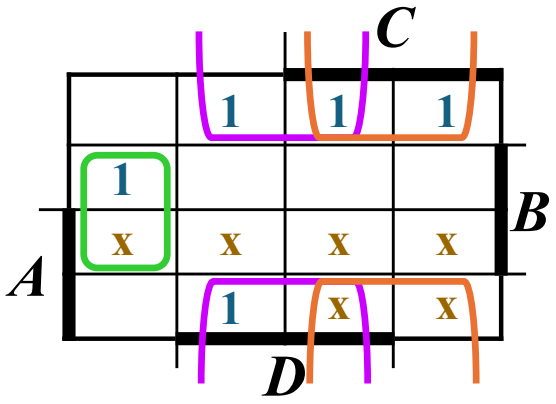
Design Procedure

- BCD-to-Excess 3 Converter

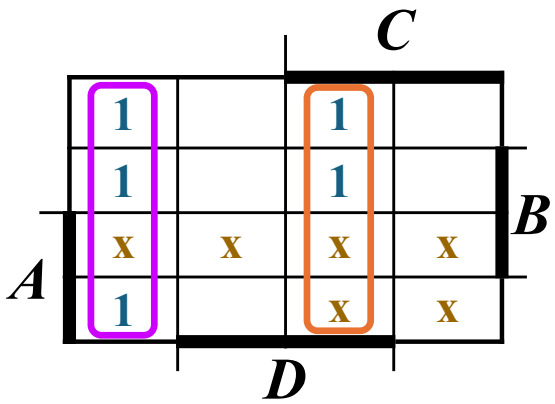
A	B	C	D	w	x	y	z
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0
1	0	1	0	x	x	x	x
1	0	1	1	x	x	x	x
1	1	0	0	x	x	x	x
1	1	0	1	x	x	x	x
1	1	1	0	x	x	x	x
1	1	1	1	x	x	x	x



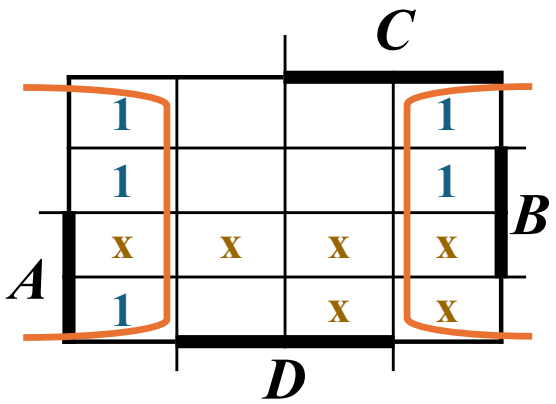
$w = A+BC+BD$



$x = B'C+B'D+BC'D'$



$y = C'D'+CD$

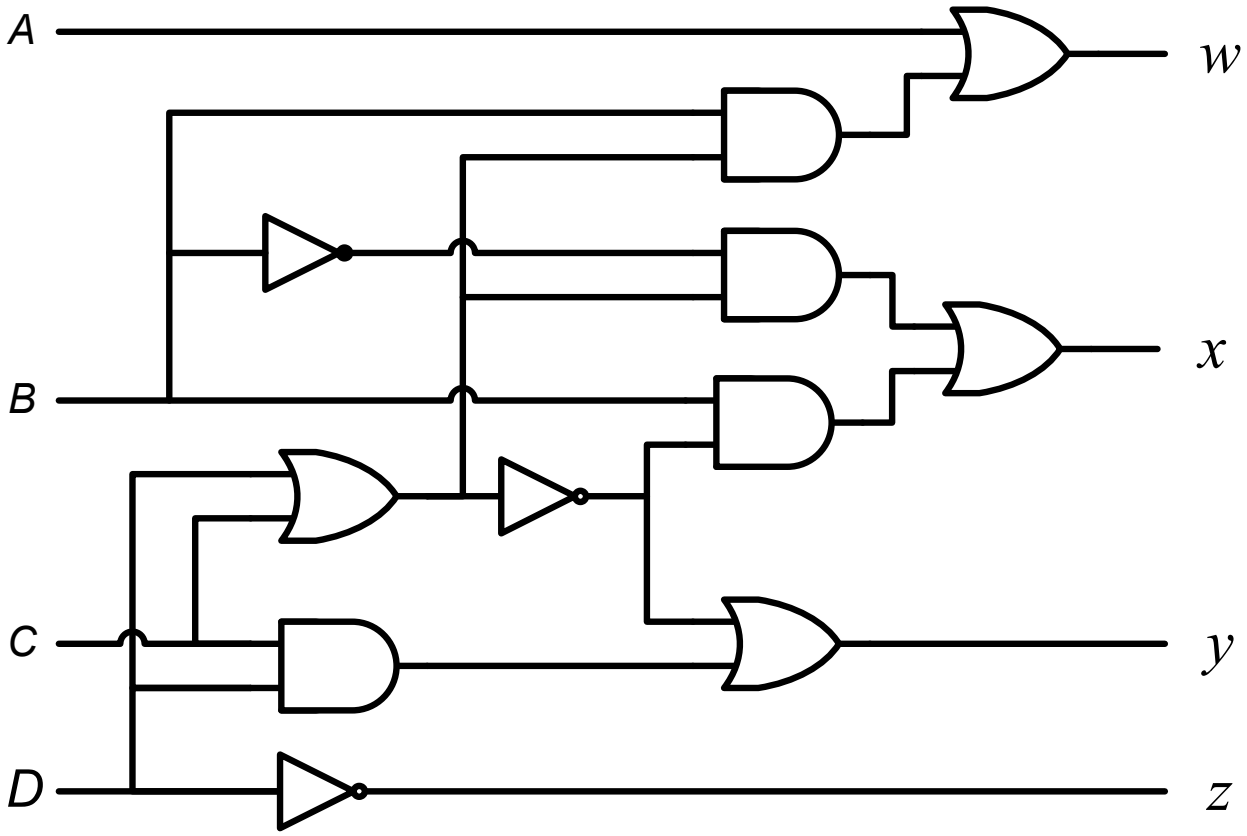


$z = D'$

Design Procedure

- BCD-to-Excess 3 Converter

A	B	C	D	w	x	y	z
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0
1	0	1	0	x	x	x	x
1	0	1	1	x	x	x	x
1	1	0	0	x	x	x	x
1	1	0	1	x	x	x	x
1	1	1	0	x	x	x	x
1	1	1	1	x	x	x	x



$w = A + B(C+D)$

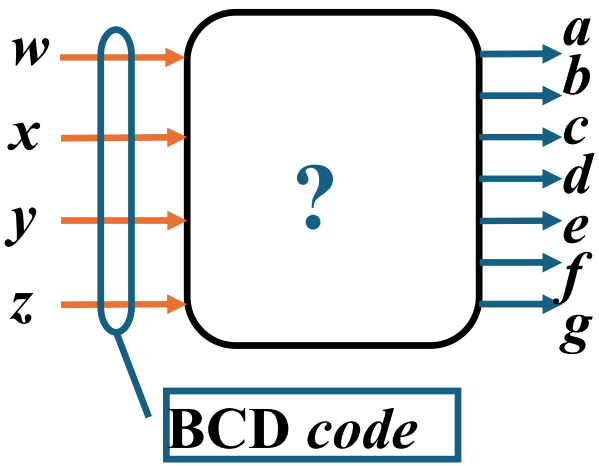
$x = B'(C+D) + B(C+D)'$

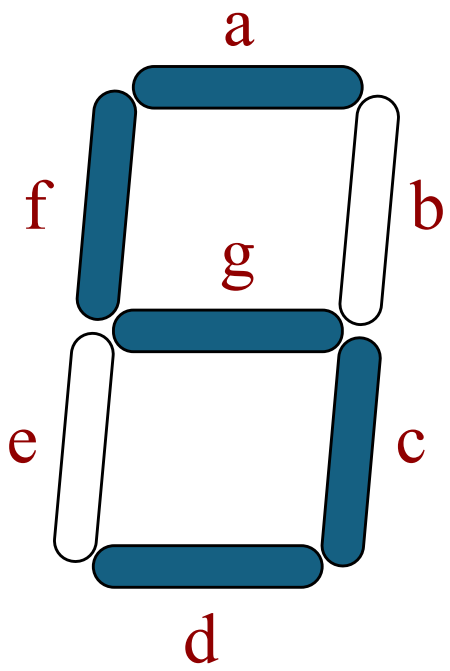
$y = (C+D)' + CD$

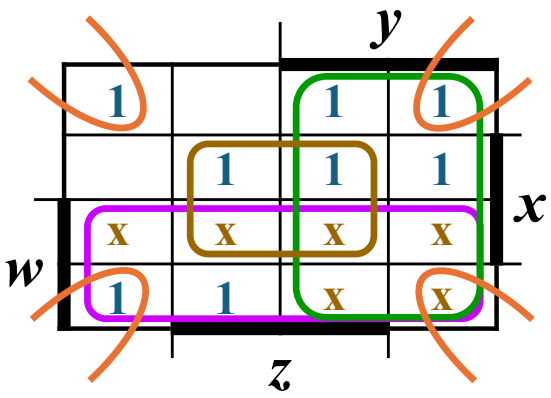
$z = D'$

Seven-Segment Decoder

w x y z	a b c d e f g
0 0 0 0	1 1 1 1 1 1 0
0 0 0 1	0 1 1 0 0 0 0
0 0 1 0	1 1 0 1 1 0 1
0 0 1 1	1 1 1 1 0 0 1
0 1 0 0	0 1 1 0 0 1 1
0 1 0 1	1 0 1 1 0 1 1
0 1 1 0	1 0 1 1 1 1 1
0 1 1 1	1 1 1 0 0 0 0
1 0 0 0	1 1 1 1 1 1 1
1 0 0 1	1 1 1 1 0 1 1
1 0 1 0	x x x x x x x
1 0 1 1	x x x x x x x
1 1 0 0	x x x x x x x
1 1 0 1	x x x x x x x
1 1 1 0	x x x x x x x
1 1 1 1	x x x x x x x





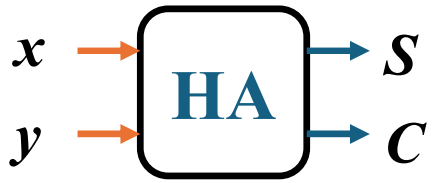


$a = w + y + xz + x'z'$

$b = \ldots$
$c = \ldots$
$d = \ldots$

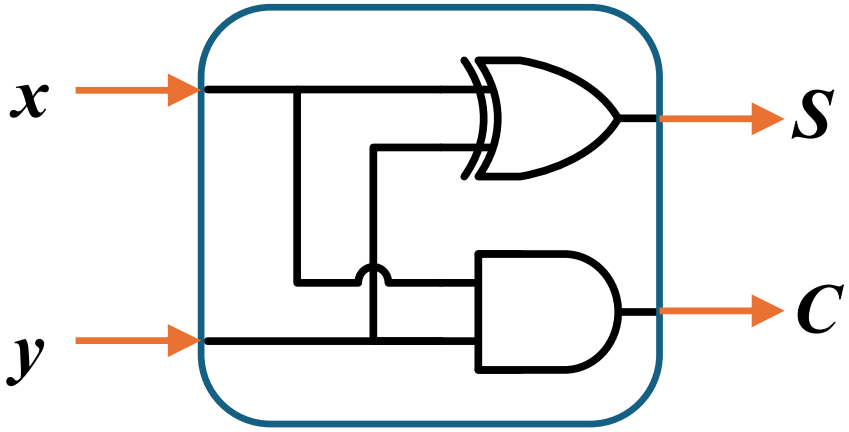
Binary Adder

- Half Adder
  - Adds 1-bit plus 1-bit
  - Produces Sum and Carry



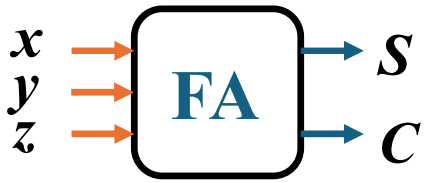
x	y	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

$$\begin{array}{rr} & x \\ + & y \\ \hline & C\ \ S \end{array}$$



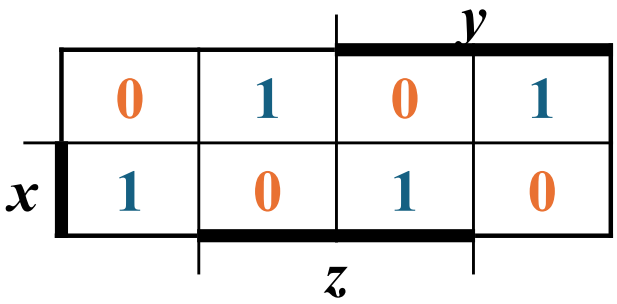
Binary Adder

- Full Adder
  - Adds 1-bit plus 1-bit plus 1-bit
  - Produces Sum and Carry

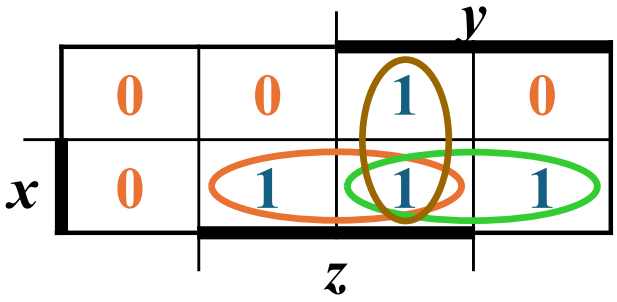


x y z	C S
0 0 0	0 0
0 0 1	0 1
0 1 0	0 1
0 1 1	1 0
1 0 0	0 1
1 0 1	1 0
1 1 0	1 0
1 1 1	1 1

$$\begin{array}{rr} & x \\ + & y \\ + & z \\ \hline & C\ \ S \end{array}$$



$$S = xy'z' + x'yz' + x'y'z + xyz = x \oplus y \oplus z$$



$$C = xy + xz + yz$$

Binary Adder

- Full Adder

$$S = xy'z' + x'yz' + x'y'z + xyz = x \oplus y \oplus z$$

$$C = xy + xz + yz$$